Università di Pisa
Department of Physics
Master's Degree in Physics

Anomalous Transport in Soaring Flights

Collective Strategies and Intermittent Search Models

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Chapter 1

Introduction

Chapter 2

The dataset

2.1 Source

The flights come from the *Coupe Fédérale de Distance* (CFD), the cross-country competition of the French free-flight federation (FFVL). Pilots declare their flights online, and each entry carries the flight's metadata and, in most cases, its GPS track. Paragliders and hang gliders have separate archives; a larger share of hang-glider entries has no track.

A season runs from 1 September to 31 August. We use the paraglider seasons from 1999–2000 and the hang-glider seasons from 2001–2002, up to 2025–2026. For each season we download the list of declared flights, kept verbatim as provenance, and every track it links. A saved list is never replaced, so each one is a snapshot of the day it was downloaded: the 2025–2026 lists were saved around the end of June 2026 and miss the rest of that season.

2.2 Tracks and metadata

Tracks are IGC files [1], plain-text records of the flight. Each `B` record is one *fix*: UTC time, latitude and longitude, a validity flag, and two altitudes, the pressure altitude from the barometer and the GNSS altitude from the satellite fix. The sampling interval depends on the recorder and need not be constant within a flight. Some recorders log no pressure altitude.

The metadata give the date, the pilot, the take-off and landing sites, the wing model and class, the scored distance and the scoring category of the declared route (free distance, out-and-return, flat or FAI triangle and a few others). The category describes the declared route only. A few paraglider entries have an incomplete date, such as `0000-00-00`, which excludes them from any comparison by date.

2.3 Glider categories

Wing design changes speed, glide performance and turning geometry, so paragliders and hang gliders are analysed separately [2]. A paraglider is a flexible canopy; a hang glider has a framed wing, flexible or rigid.

The wing class follows a different system in each discipline. Paraglider classes are those of the EN 926-2 standard, A to D, written with their equivalent in the older LTF scale ("C ou 2" reads "C or 2"), plus the CIVL Competition Class and a few other labels, such as tandem or uncertified wings. EN classes rank passive safety and the skill needed to recover from a collapse, and say nothing direct about speed or glide ratio [3]. Hang-glider classes are those of

Table 2.1. Paraglider flights per season.

Season	Flights	With track	Downloaded	Downloaded (% of tracks)
1999–2000	10	10	10	100.0
2000–2001	832	1	1	100.0
2001–2002	1,150	3	3	100.0
2002–2003	1,996	123	123	100.0
2003–2004	1,470	337	337	100.0
2004–2005	1,765	574	574	100.0
2005–2006	2,162	894	894	100.0
2006–2007	2,386	1,016	1,016	100.0
2007–2008	3,081	1,667	1,667	100.0
2008–2009	3,974	2,614	2,613	100.0
2009–2010	4,171	3,059	3,059	100.0
2010–2011	4,872	4,185	4,169	99.6
2011–2012	6,744	6,075	6,065	99.8
2012–2013	7,873	7,106	7,075	99.6
2013–2014	9,804	9,157	9,127	99.7
2014–2015	13,793	13,293	13,254	99.7
2015–2016	13,982	13,703	13,696	99.9
2016–2017	15,760	15,534	15,528	100.0
2017–2018	14,917	14,682	14,676	100.0
2018–2019	16,702	16,517	16,508	99.9
2019–2020	10,966	10,868	10,863	100.0
2020–2021	9,500	9,483	9,480	100.0
2021–2022	14,552	14,507	14,502	100.0
2022–2023	11,590	11,561	11,560	100.0
2023–2024	10,372	10,362	10,360	100.0
2024–2025	10,906	10,885	10,884	100.0
2025–2026	8,013	8,009	8,008	100.0
Total	203,343	186,225	186,052	99.9

Table 2.2. Hang-glider flights per season.

Season	Flights	With track	Downloaded	Downloaded (% of tracks)
2001–2002	248	0	0	–
2002–2003	281	6	6	100.0
2003–2004	256	28	25	89.3
2004–2005	244	35	25	71.4
2005–2006	365	80	76	95.0
2006–2007	337	100	96	96.0
2007–2008	388	136	132	97.1
2008–2009	344	155	155	100.0
2009–2010	324	136	136	100.0
2010–2011	425	147	147	100.0
2011–2012	117	103	103	100.0
2012–2013	428	406	406	100.0
2013–2014	423	409	409	100.0
2014–2015	628	595	591	99.3
2015–2016	417	413	413	100.0
2016–2017	468	462	462	100.0
2017–2018	342	331	331	100.0
2018–2019	604	594	594	100.0
2019–2020	378	373	373	100.0
2020–2021	328	328	328	100.0
2021–2022	570	568	568	100.0
2022–2023	376	374	374	100.0
2023–2024	437	437	436	99.8
2024–2025	353	352	352	100.0
2025–2026	178	178	178	100.0
Total	9,259	6,746	6,716	99.6

the FAI [4]: Class 1, with its Sport sub-class, for wings controlled by weight shift alone, and Classes 2 and 5 for rigid wings with movable control surfaces.

Chapter 3

Trajectory pre-processing

The pre-processing turns each IGC file into a trajectory fit for analysis. This chapter covers its first two stages: reading the fixes and choosing the altitude channel.

3.1 Reading the fixes

Each `B` record gives a fix $(t_i, \varphi_i, \lambda_i)$, with latitude and longitude in degrees, together with the validity flag and the two altitudes h_i^{baro} and h_i^{GNSS}. A record that does not follow the fixed layout of the format, or whose time of day or position is impossible, is discarded. A missing or unreadable altitude, including the zero the format uses for a missing reading, is marked as missing, and the fix keeps its time and position.

Time is measured in seconds from the first fix. Records carry only the time of day, so a flight crossing UTC midnight would jump back by a day; a day is added when the clock passes from the last hour of one day to the first hour of the next. Other backward steps and repeated times are left in place, as are well-formed but implausible fixes such as position spikes: the parser discards only what cannot be read. The validity flag, which marks fixes without a three-dimensional GNSS solution, is kept but not used at this stage.

3.2 Choice of altitude channel

Some recorders log no pressure altitude, so we use GNSS altitude for every flight, $z = h^{\text{GNSS}}$. Its vertical datum is not uniform across recorders: the IGC specification prescribes height above the WGS84 ellipsoid [1], the CIVL specification for hang gliders and paragliders height above the geoid [5]. A constant offset cancels in altitude increments, but comparing absolute altitudes across flights would need a datum audit.

Let N be the number of fixes of a flight and P the set of those with a GNSS altitude. The flight is kept only if

$$\frac{|P|}{N} \geq 0.95 \qquad \text{and} \qquad \max_{i \in P} h_i^{\text{GNSS}} - \min_{i \in P} h_i^{\text{GNSS}} \geq 30\,\text{m}. \tag{3.1}$$

The first condition rejects incomplete channels; the second rejects channels stuck at one value, and flights with almost no vertical excursion. Fixes flagged as lacking a three-dimensional solution count as present if they carry an altitude. Pressure altitude is never substituted for a missing GNSS value.

The barometer is checked with the same two conditions and thresholds. A flight whose barometer passes is marked, so that later cleaning steps can use pressure altitude as independent evidence; the mark does not decide whether the flight is kept.

Bibliography

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[5] Fédération Aéronautique Internationale, CIVL. Sporting code section 7h: Civl flight recorder specification. 2024 edition, stated effective date 1 May 2024, 2024. Section 3.2.1 specifies GNSS altitude relative to the geoid. The available document retains draft annotations. https://fai.org/sites/default/files/civl/documents/sporting_code_s7_h_-_civl_flight_recorder_specification_2024.pdf.